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Problem Chosen

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2003 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

Given that the arrived time of a passenger satisfies with normal distribution, we first establish a model to estimate the maximum number of EDSs required directly. And then our result is : airport A needs 35 EDSs, and airport B needs 38 EDSs.

In subsequent, for the convenience of analysis, we first introduce the notion of ‘normalized flight’. Secondly, synthetically analyze the relationship among the three curves:

the curve of the checked bags arrived;

the curve of the bags screened ;

the curve of the bags which must be screened over to guarantee flights to take off regularly.

And then we obtain an optimum rule of flight arrangement and design a ‘man-tower’ algorithm to schedule the departure of different types of flights. Thus, an accurate schedule and a preferable number of EDSs can be given: the schedule is shown in our paper; the number of EDSs airport A needs is 18, and that of airport B is 19.

Further more, in order to improve security, the ETD machine is introduced. By modifying the parameters in the EDS model, the number of ETDs can be obtained quickly, and the corresponding problems given in task 7 are discussed too. And the number of ETDs required is 12 both in airport A and B.

At last, under other practical conditions, another model is given to estimate the number of EDSs.

Security Screening at Airport

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1.Problem Restatement

To guarantee the security of passengers, the TSA has been actively purchasing EDSs and deploying them at airports throughout the nation. However, the manufactures are not able to produce the expected number of EDSs required. Therefore a model is required to estimate the correct number of devices to

deploy at each airport. Meanwhile the security screening takes time and might delay passengers, so a new schedule should be determined. Further more, because of the introduce of ETD, their number and if the schedules need to be changed should be discussed.

2.Problem Analysis

Manufacturers and airlines hope that there is an expected number of EDSs required. The bigger the number is, the more security can be guaranteed. However, given the limited space and funds available for each airport, we should design a rational model to estimate a correct number. The number is related to different factors as the following:

- a. The passengers flow, which is the vital important.
- b. The checked bags flow, which is determined by the passenger flow.
- c. The ability of the machine screens the checked bags.

In subsequent section, because the security screening takes time and might delay passengers, so a new schedule should be determined. Based on the model that estimates the number of EDSs, we use an algorithm named as “man-tower” algorithm to give an accurate schedule.

Further more, because of the introduction of ETDs, we modify our model to incorporate the use of ETD machines. And in this section, the number of ETDs and the schedule are discussed. Also, we consider the cost of the introduce of ETD and if the ETDs can replace the EDSs from the view of fund.

At last, we examine the possible effect of changes in device technology, cost, accuracy, speed, and operational reliability and give our conclusion. And we do more analysis to estimate the number of EDSs under another condition.

3.Notations

v the speed EDSs screen at, and its unit is bags per hour

N the number of EDSs

M the number of ETDs

R the number of the passengers arrived at the airport

B the number of checked bags arrived at the airport

B_r the number of checked bags screened in reality

T_1 the time interval between the stop time of screening and the time of takeoff, which equals to 30 minutes in our model

T_p the peak hour, which equals to 1 hour

T_0 the range of time that one passenger arrives for a given flight, which equals to 75 minutes

4.Modeling for Task 1 and 3

Outline

To complete the objects in task1 and task 3, that is to estimate the number of EDSs required at two of the largest facilities in the Midwest Region and to give an accurate schedule, we establish the models as the following three steps:

- ①Make assumptions and preparations. This is the base of the following steps.

②Estimation of the number of EDSs required. First, we give a maximum number; Second, we give an acceptable number.

③Consider the reality, that is takeoff occurs around the peak hour. Under this case, basing on step2 and modifying the irrational factors in the model of step 2, we get a better and more practical model.

4.1 assumptions and preparations

Assumptions

- 1) Peak hour is one hour, which can be given from the data in the Appendix A in the problem.
- 2) Every airport has at least 2 runways.
- 3) 20% of the passengers do not check any luggage, 20% check one bag, and the remaining passengers check two bags.

Instruction

- 1.It is beneficial to deal with the off time of EDS/ETD equivalent to fall down their efficiency in some extent, for example, the operational time of EDS is 92%of the time, that is $v' = 92\% \cdot v$, where v' is effective velocity, and v is the ideal velocity.
- 2.The peak hour of flights have influence on the peak hour of checked bags flow, and the latter is advanced to the former. In our model, the lead is 82.5 minutes.
- 3.The passengers flow can be converted to the checked bags flow, which is described in appendix 1.

Preparations

the number of passengers flow

To estimate the number of EDSs, we should first estimate the total number of checked bags. To estimate the bags, we should find out the total number of passengers arrived at any time.

Generally, passengers' arrival is random. Meanwhile their arrival is influenced by many factors subjectively and objectively. Thus, we deal with the probability of every passengers flow submitting to normal distribution, or

$T \sim N(\mu, \sigma^2)$, that is

$$\varphi(t; \mu, \sigma) = \frac{1}{\sigma \cdot \sqrt{2\pi}} \exp\left\{-\frac{(t - \mu)^2}{2\sigma^2}\right\} \quad (1)$$

In this section, the parameter μ and σ should be determined.

Given that passengers typically arrive for their flight between forty-five minutes and two hours prior to their scheduled departure time in the problem, we get the range of time that one passenger arrives for a given flight is 75 minutes. Using the symmetry of normal distribution, it's rational to set μ as 75/2 minutes, and then $\mu = 75/2$ minutes.

Because the event that passengers arrive for a given flight before 2 hours or after 45 minutes hardly occurs. Applying the 3σ principle,

$$|T_0 - \mu| \geq 3\sigma \quad (2)$$

Where, $T_0 = 75$ minutes, $\mu = 75/2$ minutes

Then $\sigma \leq 12.5$ minutes

To determine the shape of normal distribution curve, we take $\sigma = 12.5$ minutes.

Thus,

$$\varphi(t; \mu, \sigma) = \frac{1}{12.5 \cdot \sqrt{2\pi}} \exp \left\{ - \left(t - \frac{75}{2} \right)^2 / 50 \right\} \quad (3)$$

And the normal distribution curve is shown in figure 1 as follows:

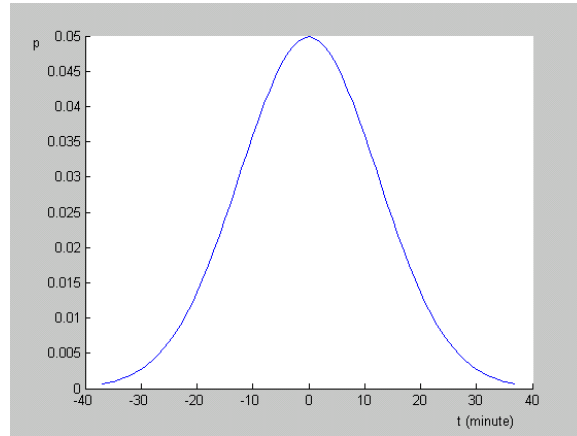


Figure 1 the normal distribution curve

Let $t_0 \in [0, T_0]$ and Δt is infinitely small, then during $[t_0, t_0 + \Delta t]$, we can get the average number of passengers flow for a given flight as:

$$E(t_0) = \sum_i^M N_i \cdot \varphi(t_0) \cdot \Delta t \quad (4)$$

Where, $N_i = \begin{cases} 1, & t_0 < t < t_0 + \Delta t \\ 0, & \text{others} \end{cases}$, and M is referred to the total passenger number of a given flight.

$$\text{Then } E(t_0) = M \cdot N_i \cdot f(t_0) \cdot \Delta t \quad (5)$$

Because the value of N_i is independent to i , when Δt and M is fixed,

$$E(t_0) \propto f(t_0) \quad (6)$$

Hence, we have shown that the shape of this curve about $E(t_0)$ is the same to that of the normal distribution curve in figure 1. Figure 2 shows this curve. The passengers a given flight holds determine the area below the curve, and the departure time of the plane determines the middle position in the Cartesian coordinates.

From figure 2, we can get the average number of passenger flow for a flight at an arbitrary time. Taking the same

method, the other flights' can be easily obtained. That is every flight has its corresponding curve. And generally, superimposing the average number of passenger flow of all the flights, we can get a new curve.

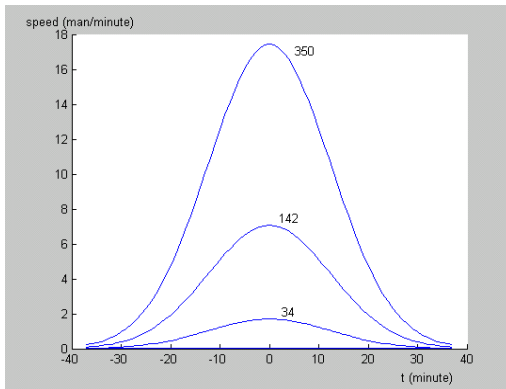


Figure 2 the average number of the passenger flow

At this section, we are devoted to determine the specific shape of the average number of passenger flow of all the flights. According to assumption 1 in Task 1, the result of superposition is shown in figure3. We can see that, the superposition occurs the same time. Thus, the maximum number of EDS required can be estimated in the following text.

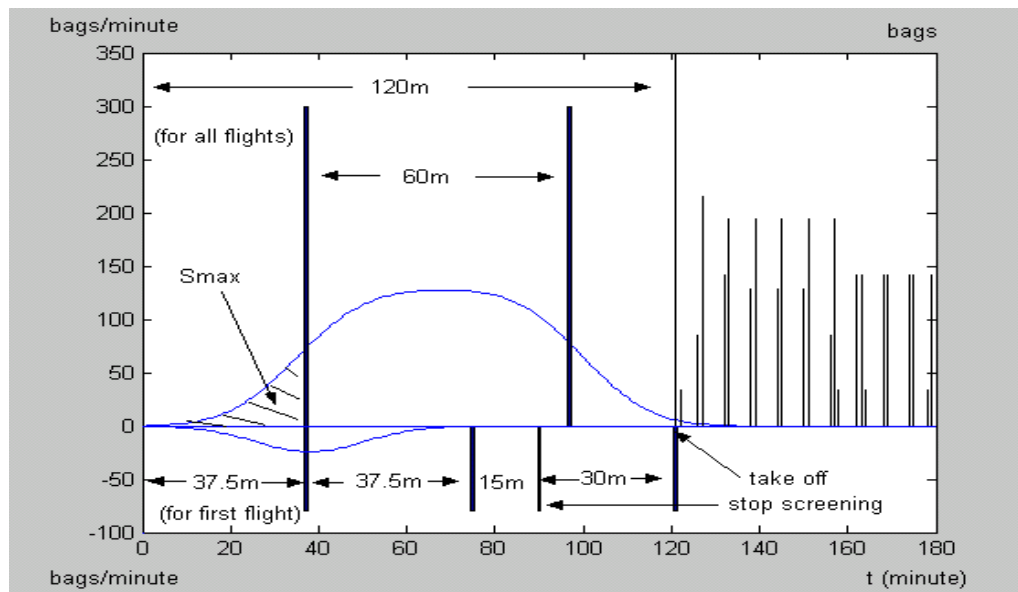


Figure 3 the checked bags at an arbitrary time

4.2 Estimation

Estimation of the maximum number

In this section, we need some assumptions more as follows:

1. All the planes take off at the same time. Though, it is impossible in fact. We make this assumption only to find the maximum number of EDSs that is needed. It is the same to the assumption below.
2. All the flights fly with 100% of their seats occupied.
3. Once passengers arrive, their checked bags can be screened until before the 30 minutes of takeoff.

As depicted in figure 3, we get the total number of checked bags flow at any time and we use l_1 to express this curve.

In the following part, we focus on the number of EDSs required at two of the largest facilities in the Midwest Region.

The ability of all the EDS screening bags at arbitrary time should be discussed. It is related to the total number of EDS and the screening speed. Given that each device can examine between 160 and 210 bags per hour by the experimentation, we take its average speed 185 bags per hour [$(160+210)/2$], referred to as v . And the total numbers of EDS required is defined to N . So we get the number of checked bags screened in reality as follows:

$$B_r = N \cdot v' \cdot t \quad (7)$$

Under the condition that all the checked bags should be screened ahead of 30 minutes and the bags screened is less than the bags arrived (Seeing from the figure 4, that is , the curve of B_r referred to l_2 has to cross the point A, and l_2 is below l_1) , the maximal slope of any point in l_2 can be obtained, which is the biggest ability of EDSs screening, and let k is the slope.

Thus, we can estimate the maximum number of EDSs required, $N_{\max} = \frac{k}{v'}$. Then, A airport needs 36, and B airport needs 38.

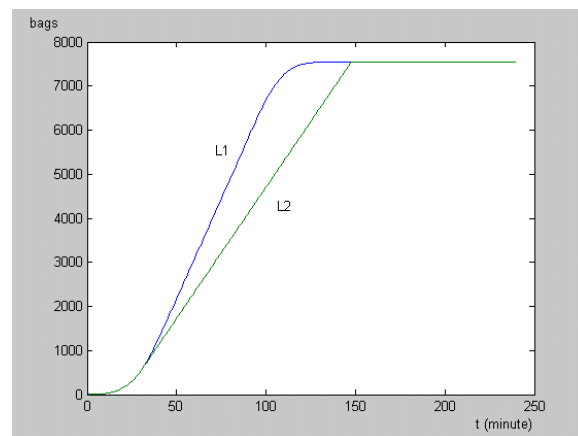


Figure 4 the bags arrived and the bags screened

Estimation of an acceptable number

To estimate an acceptable number, we modify the assumptions to:

Consider the seats occupied in reality, as the problem suggests.

The flights take off in an equal interval, and the subsequence of takeoff is random distribution.

In an exactly analogous manner above, we can get :

The acceptable number of EDSs A airport needs is 16, and B airport needs is 18.

4.3 A precise model

Further more, based on the model above, we establish a more precise model to estimate the number of EDSs and the schedule required.

At first, the assumptions needed in the following should be explained:

a. We assume that the number of flights around the peak hour is obviously decreased. Let the peak time last the time of T_p , then the number of flights after peak hour is less as $T_p / (52.5 + T_p)$ times as the average number in peak hour, where the equation will be explained in the following.

b. Compared to the last time of the peak hour, the time interval of the two peaks is long enough.

Secondly, modeling as follows:

1. Normalized Flight

For the convenience of analysis, we convert the real flights to the normalized flights with the same ability of carrying passengers. Therefore, in the following model, only relatively small and homogenous flights should be considered. A normalized flight may be consisted by several different types of flights. We have no common standard about the size of the normalized flights, but if the size is too big, that more other types of flights needs is definite. Meanwhile the flights almost take off at the same time, their number is limited by the number of runways.

The detail explanation about the size and the number of the normalized flights is discussed in the appendix 3.

2. Analyzing

①The meaning of curve l_1 (the bags arrived) and l_2 (the bags screened) has been explained in 4.2, and the arbitrary point in l_3 means that the checked bags should be screened up. The three curves have been depicted in figure 5. The screening of checked bags must end up ahead of 30 minutes when the corresponding flight takes off, thus shape

of the curve l_3 depends on the case of the takeoff in the peak hour. Specifically, if the flight takes off in equally spaced time, then the shape is a straight line. Superposing the bags flow curves and integrating, l_1 will be bumped towards the direction the ordinate increases, otherwise l_3 will generate distortion.

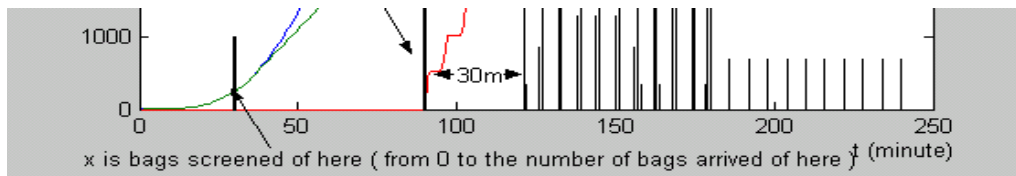


Figure 5 the bags arrived, the bags screened and the bags must be screened

②The object is to find the minimal number of EDSs required. From the figure 5, that is to make the maximum slope of any point in l_2 minimize, and we call the result value minmax. Noticed that, the distortion of l_3 and the lower of l_1 , both of which cause the minmax slope to increase. However, the upper of l_1 has no influence on the minmax. Thus, the conclusion is that departure in equally spaced time is the takeoff manner determined.

3.conclusion and analysis

(1)As shown in figure 5, the minimal number of EDSs is

$$N_{\min}(x) = \frac{B - x}{\left(120 - 75 - T_1 + \frac{T}{2} + T_p\right) \cdot v'} \quad (8)$$

Where, x is the total number of bags arrived which has been screened up at the beginning time of the checked bags peak hour. The value of x is related to the passengers flow in the previous one hour.

Considering the two extreme conditions: one is, the passenger flow is big in the previous one hour, and there is no time to screen these bags, thus $x = 0$; the other is, if the passenger flow is small, and there is enough time to screen up all these bags, then $x = S_{\max}$, where $S_{\max}$ is the area of the shade part in figure 3. That is,

$$N_{\min}(S_{\max}) \leq N_{\min}(x) \leq N_{\min}(0) \quad (9)$$

And $N_{\min}(0)/N_{\min}(x_{\max}) = x_{\max}/B$, the value of which is small. Considering the peak hour as one hour, approximately 1/8, if the $N_{\min}$ is 20, and then the error of the result won't be over 3. If the peak hour is more than 1 hour, the value will be smaller. Synthetically considering the two cases above, it is adequate to use $N_{\min}(0)$ as estimation value.

For the peak hour is less than 1 hour, the result depends.

(2)Applying the algorithm about the normalized flight in appendix 3, we get the schedule under the condition above. Table 1 and 2 and figure 6 shows the result.

And the preferable number of EDSs is

$$N_A = 17, N_B = 18 \quad (10)$$

The explanation to the table below, take the first two data for example, that is a flight with 85 seats take off at the time of 1 minute in the peak hour.

Table 1 the schedule of airport A

Departure time	Flights capacity	Departure time	Flights capacity	Departure time	Flights capacity
1 85		24 128		44 34	
1 350		25 142		48 46	
2 34		25 194		48 142	
2 34		30 34		49 142	
6 34		30 128		49 142	
6 85		31 142		54 46	
7 142		31 194		54 142	
7 215		36 46		55 142	
12 142		36 85		55 142	
13 142		37 142		58 34	
13 194		37 194		59 34	
18 34		38 34		59 142	
18 128		42 46		60 142	
19 142		42 142		60 142	
19 194		43 142			
24 34		43 142			

Table 2 the schedule of airport B

Departure time	Flights capacity	Departure time	Flights capacity	Departure time	Flights capacity
1 85		20 34		42 128	
1 350		24 34		43 142	
2 85		24 85		43 194	
6 85		25 194		44 34	
6 85		25 194		48 46	
7 142		26 34		48 128	
7 215		30 34		49 142	
12 46		30 142		49 194	
12 85		31 142		54 46	
13 142		31 194		54 128	
13 215		36 46		55 142	
14 34		36 128		55 194	
18 34		37 142		59 46	
18 85		37 194		59 128	
19 194		38 34		60 142	
19 194		42 46		60 194	

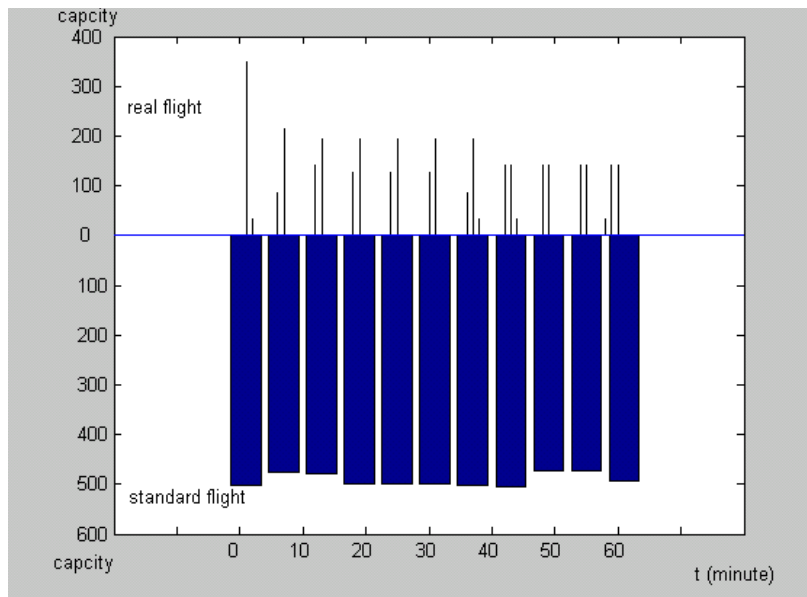


Figure 6 the departure of the real flight converted to the normalized flight

5.Task2、 4 and 6

To keep the integrity of text structure, we put the position paper, recommendation and memo in the end in chapter 11.

6. Task 7

6.1 Model modifying

In this part, we modify our model established in chapter 4 (to be more precise, in chapter 4.3), to get some results we concern.

the number of ETDs

Recall how we estimate the number of EDSs in chapter 4, considering the operational time and the screening speed, 1EDS equals to g ETD in the ability of screening bags per hour, where g can be obtained from the equation below:

$$185 \cdot 92\% = 45 \cdot 98\% \cdot g \quad (11)$$

Then, $g = 3.86$

AS given in the problem that, up to 20% of the passengers will need to have all their checked bags screened through both an EDS and an ETD, and 80% of the passengers have checked bags. That is $80\% \cdot 20\% = 16\%$ bags need to be checked by both EDS and ETD, then the number M of ETDs required is:

$$M_A = g \cdot N_A \cdot 16\% \quad (12)$$

$$M_B = g \cdot N_B \cdot 16\% \quad (13)$$

So according to our EDS model, we obtain $M_A=12, M_B=12$.

Schedule

Since the screening by ETD will takes more time, which may cause some delays. If the 20% of the passengers all miss their flights by m minutes, there are three ways to solute this problem:

- i the plane take off behind schedule by m minutes;
- ii airlines claim that all the passengers arrive m minutes earlier, where m depends on the delays due to the screening by ETD;
- iii add the number of EDSs, and pick up the total speed of screening by EDS, because the screening by EDS is prior to the ETD.

Value and Cost

Let each EDS/ETD needs one person to operate, and the operating cost of EDS is \$10/h per person, that is \$36,000per year per person, then in n years increasing one ETD needs to cost:

$$C_t = 45000 + C_i + 10 \cdot 36000 \cdot n \quad (14)$$

Where, C_i is the cost of installing, exactly, $C_A = \$100,000$, and $C_B = \$80,000$.

Then, the introduction of one ETD needs \$505,000 and \$485,000, respectively in airport A and B in the first year. As A airport needs 12 ETD, the cost in the n th year is $1,740,000 + 1,312,381 \cdot n$, B airport needs 12, and its cost in the n th year is $55,071,429 + 4,050,000 \cdot n$.

Should the ETDs replace any of the EDS devices?

Given that k ETDs replace one EDS, and then in n years, we can save the cost as:

$$S = C_{ETD} - C_{EDS} = (1000,000 + C_i + 360,000) - g \cdot (45,000 + C_i + 360,000 \cdot n)$$

Simplify the equation above,

$$S = 1000,000 - 45,000 \cdot g - (360,000 \cdot g \cdot n - 36,000 \cdot n) \quad (15)$$

Investigating the equation above, when $n \geq 0.6$, $s < 0$. That is, in a short time, exactly before July of the first year, the measure that ETD replaces EDS can save certain fund for the airlines. But in the long run, it's inadvisable.

Thus, our conclusion is that the ETDs can't replace any of the EDS.

6.2 Analysis of accuracy

The accuracy is the percentage of the checked bags screened correctly occupying the total checked bags. Exactly, as far as one EDS, if there are 100 bags to be screened, the most possible is that 1.5 bags have been misjudged. That is, if there exists explosive devices in these bags, then it may be taken for the bags with no explosive devices, and vice visa.

If the result of screening by EDS and ETD is consistent, the accuracy can reach 99.9955%, which can be obtained from the expression $1 - (1 - 98.5\%) \cdot (1 - 99.7\%)$. In the circumstances, the probability of the bags with explosive devices passed is 0.0045%. That is very small, so we consider the result of screening is accurate.

If the result of screening by EDS and ETD is inconsistent, the more screening in detail should be done. We may check out some bags with higher risks, the number of which is very small, approximately $20\% \cdot (1.5\% + 0.3\% - 1.5\% \cdot 0.3\%) = 0.36\%$. Since the screening of bags with no risk takes time, thus the schedule of flights has to be changed. But

given that the inspectors are expert and experienced, then the extra time is relatively small. To guarantee all the passengers not to miss their flights, all the corresponding flight should take off a little behind schedule. Otherwise, the number of EDSs should be increased and we can get the corresponding number from the equation (8) .

6.3 Memo

(See chapter 11 Memo 2)

7.Task 8

7.1 Sensitivity analysis

Let's do some sensitivity analysis in our model. If the speed of EDS increases by 5 percentage point, that is, from 185/h to 194.25/h in average, then the number of EDS needed will decrease by 1, for both airport A & B. If the operational reliability of EDS increase by 5 percentage point, that is from 92% to 97%, then the number of EDS needed will decrease by 1, for airport A & B both, too. These conclusions indicate how we will benefit from the improvement of technology.

7.2 Recommendation

In our model, we overpass the influence that the bags with no risk misjudged. So the model can't reflect the economic influence of security screening. In fact, the increasing of accuracy is good for economy, for example, the reducing of the delays and less labor needed to do desk checking.

For the existing safe system consisted of EDS and ETD, the security can't be guaranteed absolutely. Because both EDS and ETD are emerging technologies, especially to the ETD, which has not been federally certified, as the maturity of production technology and the progress of the software and hardware, the speed of screening will improves increasingly, so does the accuracy. So it is worth enlarging the investment in the science 、 technology and engineering.

In addition, as the new technology develops, some new explosive devices that can avoid the screening may be emerged. Thus, the security equipment should be refreshed in time. Therefore the state should invested funds to the elementary study fields.

8. Estimation of the number of EDS under another condition

8.1 Model under the consideration of one hour

First, we made two assumptions:

- 1.Both A and B airports have two runways, which means at most 2 planes take off at the same time.
- 2.The minimal interval two adjacent flights take off is 1 minute.

The curve of total checked baggage flow can be easily obtained by superpose the curves partially shown in figure 2 and is then shown in figure 7. An arbitrary point in this curve denotes the velocity of the arriving bags at the time. For the convenience of analysis, without proof, we assume that the minimal speed for screening bags $N \cdot v'$ is no less than the velocity h the bags arrived. Under this condition, the process of screening is real time. That is, passengers won't waste time because of screening the checked bags. In another word, the waiting time of passengers in queue is minimal.

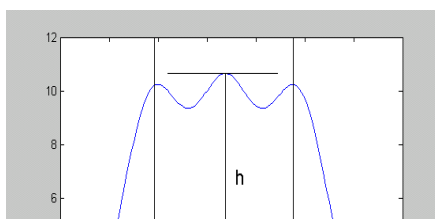


Figure 7

Notice that, the area below the curve is fixed, and the length of interval is fixed. What we should do is to find a superposition way to minimize h , and then n will be minimized ($n=h/v$). In order to let all the EDSs work at the maximum speed for a time as long as possible, the top of the curve should be as horizontal as possible.

Based on the analysis above, we design a method named “man-tower” algorithm.

Noticed that, all the planes have to take off in one hour, the horizontal distance of the bilateral curve (See figure 8, that is the horizontal distance from a to b) is less than 60 minutes.

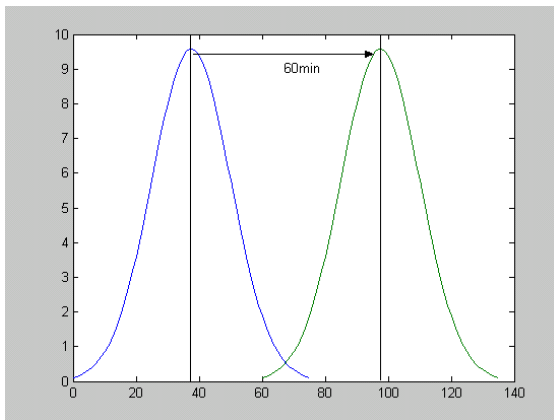


Figure 8

Applying the “man-tower” algorithm (See appendix 2) we could get the peak hour departure schedule, from which we can see that many flights centralize in a relatively small number of points. For example, at the end of the peak hour, 15 flights have to take off. That is unrealistic.

Modify this algorithm. Given that there are two runways and the interval is 1 minute. Set 60 points at the time-axis, at most 2 flights at every time point can take off. That is, only from the “man-tower” algorithm, only two men can be put at one point. Based on the algorithm above, once a point is occupied by two men, we should find another point. Thus, we can get the minimal number of EDSs required at airport A is 22, and that at airport B is 24. Meanwhile, the schedule can be obtained

8.2 Model under the consideration of several hours

In fact, there are flights around peak hour. Thus, this case should be taken into account. Here, we make two senseful assumptions:

- ① The number of flights in peak hour is proportional to that around peak hour. Therefore, the passengers their flights hold is proportional too, let c is the factor of proportionality.
- ② The curves of bags flow are bilateral symmetry in peak hour.

In the subsequent section, we focus on the bags flow in several hours. Based on the analysis in step 2, the peak hour of flights have influence on the peak hour of checked bags flow, and the latter is advanced to the former. In our model, the lead is 82.5 minutes. Use the similar way of analysis in 8.1, to guarantee the waiting time of passengers is minimal, meanwhile 100% screening, the minimal speed of screening bags $n \cdot v'$ is no less than the velocity h the bags arrived

The difference of the cases between the 8.2 model and 8.1 model is that: in the former, the checked bags flow due to the flights in peak hour can be screened before the peak hour, for example, as shown in the shade part C of figure 9. Meanwhile, some bags before the peak hour should be screened in the peak hour; otherwise the passengers will miss their flights due to screening, as shown in the shade part D. According to the assumptions ① and ②, C is proportional to D. Specifically speaking, the area of D equals to the area of C multiplied by c (here $c=80\%$).

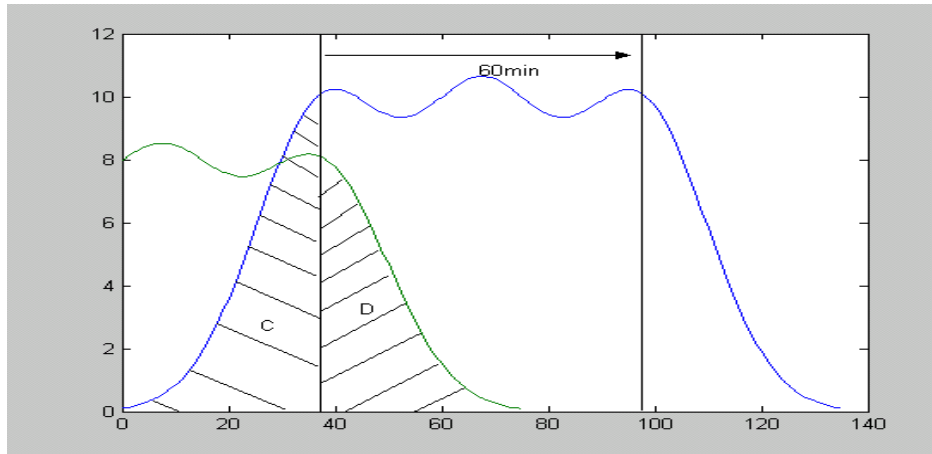


Figure 9

Take a curve of the bags flow for a given flight for example. As shown in figure 10, the peak hour of bags ranges from t_1 to t_2 . Based on the analysis above, multiply c to the shade part of l_1 outside the range $[t_1, t_2]$, then converted it to the interval $[t_1, t_2]$ symmetrically (the curve l_2). Superposing l_1 and l_2 , a new curve is obtained. Applying the “man-tower” method, the minimal number of EDSs required and the schedule can be given. For there isn’t any queue, the number of EDS should be larger than that we get from chapter 4.3. The peak hour departure schedule can be obtained, easily meanwhile.

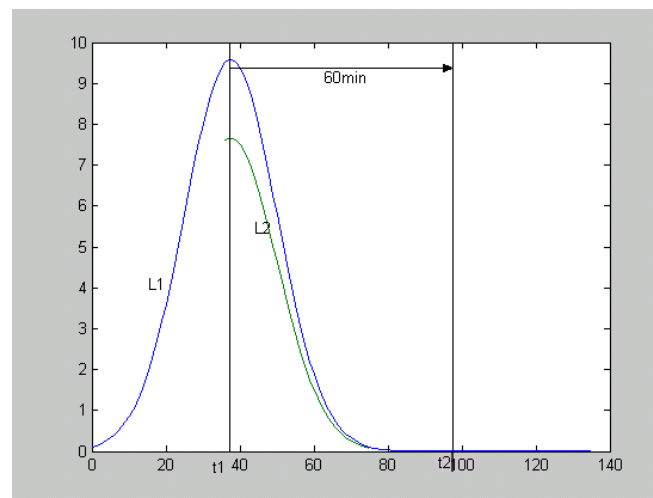


Figure 10

9.Strengths and Weaknesses

Our model is generalized and parameterized, so it's easy to apply our model to other airports.

It can produce a preferable number of the EDS or ETD required and a precise schedule.

2% of flights cancelled each day is not considered.

10.References

[1]Yu Jinghua, Shi Beiyuan and Yang Weiuan. Jan. 2001. Probability Theory and Mathematical Statistics. Publishing the Zhong Shan University Press.

[2]Ye Qixiao. Sep.2000. Mathmatical modeling. Publishing educational press of Hunan.

11.Position paper, recommendation and memo

Position paper (on security-related objectives and constraints)

In fact, we may save relatively a large amount of money if we do not purchase these devices for screening passengers' checked bags, at the cost of higher risks from the passing of explosive devices, which threaten the security of the plane, the lives of the passengers, the benefits of the US. All this, will do damages to the airlines, and these damages weigh much more than the goods the airlines get from the money saved. Thus, we think it considerably important to mandate 100% security screening. However, the balance between fund and security also becomes an important factor. So we concentrated on the finding of the minimal number of needed EDS, under the condition that all the checked baggage will be screened. And this task has been fulfilled in chapter 4 in our paper.

For the sets of flights described in Table 6 in appendix, the airlines must work within some constraints as well. Primarily, the airline must ensure that all flights take off almost accurately according to their schedule times. Second, the airline must make sure that two flights can take off simultaneously on their own runway (if the airport has two major runway). Third, the airline must guarantee that the minimal time interval between the departure-time of two consequent flights will reduce to at least one minute. Finally, the schedule of the departure of peak hour flights must be in accordance with that attained in our model.

Recommendation (to Mr. Sheldon)

Based on our analysis in chapter 4, we suggest the airline took some measures to ensure EDS to work efficiently. Specifically, if the EDS is not working at its highest speed, there will be no baggage waiting to be checked; or there will be some bags waiting, then, the bags that will be send to the earliest flight should be first picked up and send to the EDS for screening, then the bags that will be send to the second earliest flight, and so on. In a word, our recommendation is: the earlier the flight takes off, the earlier the checked baggage should be screened That means the airline should first, offer a new job picking the bags for relatively earlier flight and send them to the EDS, second, offer a new job marking all the bags for their flights when checking. The money spend on the two jobs could be ignored compared to the cost caused by the delay of flights.

Memo 1

There are several parameters in our model in chapter 4, such as T , T_1 and T_2 etc. The values of these parameters vary with different airports. However, according to the equation below (equation (8) in chapter 4.3):

$$N_{\min}(x) = \frac{B - x}{\left(120 - 75 - T_1 + \frac{T}{2} + T_p\right) \cdot v'},$$

If the peak hour is over 1 hour, the value of x tends to be 0.

The arrangement of the departure schedule will be a little more difficult to get. Generally speaking, by using the algorithm shown in appendix 3 to constitute 'normalized flights', we could get the right schedule.

Memo 2

If we require up to 20% of the passengers have all their bags screened through both an EDS and an ETD machine, the probability that explosive baggage passes will be greatly reduced. Specifically, according to the analysis in chapter 6.2, the probability that explosive baggage passes will decrease from 1.5% to 0.0045%, that is more than 300 times. However, it will cost plenty of money (for example, \$500,000 for airport A) to purchase, install and pay for the operation salary in the first year. The amount of money do not include the possible increase due to delay of flights, and will goes up as years go by.

Regardless the improvement in security, from the conclusion drawn in chapter 6.1, we can see that if one EDS is replaced by 3.86 ETD in average, which have the same processing ability to the former, we will save money in about 7 months, but will lose money after then. That is to say, EDS is more financially feasible than ETD in the long term. Of course, ETD is more accurate. The balance between fund and security should be carefully considered.

12.Appendix

12.1Appendix 1

Checked bags flow

According to the assumption 4, it is easy to transfer the passenger flow to the checked bags flow by using the following formula:

$$B = R \cdot 20\% \cdot 1 + R \cdot 60\% \cdot 2 = 1.4 \times R \quad (7)$$

Where, R is passengers arrived, and B is bags arrived. So the curve reflects the number of checked bags has the same shape with the shape that shown in figure 8, and figure 9 shows its character.

12.2 Appendix 2

man-tower algorithm

1. Calculate the area under the curve of checked baggage flow of every flight in the peak hour, we get the value of area about each flight
2. Arranging the curves mentioned above in a line from the biggest to the smallest, we call the number 1curve the biggest man, and call the last one the smallest man.
3. We first put the two biggest men to the bilateral.
4. Superpose the two men chosen 1, and get a new man (a new curve, and we call the lowest point of this curve the shoulder of the man). See figure 11.
5. Seek out the biggest one from the rest men, superposing his head (the highest point in the curve) to the shoulder of the man obtained from step 4, and then we get a new man.
6. Repeat step 3 to step 5 until all men find its position in the Cartesian coordinate system.

7. A curve of total bags flow can be obtained, and the schedule required can be given.

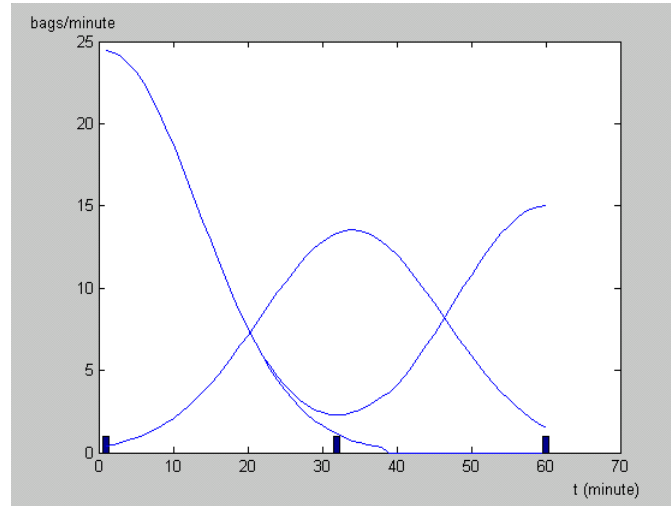


Figure 11

12.3 Appendix 3

The size of normalized flight and the algorithm on how to produce a schedule

Let the normalized flights take off in the same interval, the outline of l_2 can be straight. The real flights can be converted to normalized flights, and they take off as the same manner with normalized flight. Due to the limitation of the number of runways, if there are so many real flights to consist of a normalized flight, it's impossible to take off at the same time. Here, we can move these flights ahead or aback a little, the influence of which to the result is very small.

In addition, to make l_2 can't cross with l_1 easily, arrange the normalized flight with big flights at the fore.

On paper, the more the number of normalized flights is, the better the result is. However, it is limited by the size and the number of the flights in peak hour. To the airport A and airport B given in the problem, we get 10 normalized flights, that is the density of the normalized flights is 10 per hour. To the other small airports, we can carve up as the same method.

Algorithm:

1. Let the density of normalized flights referred as p is 10 per hour.
2. Let the number of normalized flights referred as N equals to $p * T_p$;
3. Let the size of normalized flights S equals to C/N , where C is the total size of all the flights.
4. According to the sequence from big to small, add the flight to the minimal normalized flights, then we get the real size of normalized flights S' .
5. If the error between S' and S is over 20%, then turn back to 4, otherwise stop here.
6. Let $p = p - 1$, then jump to 2.

12.4 Appendix 4

the source program

```
function init
ZT_init;


---


%data.m, to get data
H0 = [34,10,8; 46, 4,6; 85,3,7; 128,3,5; 142,19,9; 194,5,10; 215,1,2; 350,1,1,];

HA = zeros(sum(H0(:,2)),1);
HB = zeros(sum(H0(:,3)),1);

k1 = 1;
k2 = 1;
for i=1:size(H0,1)
    for j=1:H0(i,2)
        HA(k1) = H0(i,1);
        k1 = k1+1;
    end;
    for j=1:H0(i,3)
        HB(k2) = H0(i,1);
        k2 = k2+1;
    end;
end;

HA = HA*1.4;
HB = HB*1.4;
HC = ones(1,40)*85;


---


function r = allow(H,X)
X = X - 60*2 + 38;

tmp = zeros(1,60*4);
for i=1:length(H)
    tmp1 = N(-37:37,H(i));
    k = (X(i)-37):(X(i)+37);
    tmp(k) = tmp(k) + tmp1;
    k = (X(i)+37+1):(X(i)+37+15);
    tmp(k) = tmp(k)+ tmp1(75);
end;

r = zeros(1,60*4);

for i=1:length(X)
    r(X+37+16) = tmp(X+37+16);
end;
```

```

function [P0,d] = doPass(P,k)
k = k/60;%3600 * (70*60);
P0 = zeros(1,60*4);
d = zeros(1,60*4);
P0(1) = 0;
for i=2:(60*4)
    P0(i) = P0(i-1)+k;
    if (P0(i)>P(i)), P0(i) = P(i); end;
    d(i) = P(i) - P0(i);
end;

```

```

function X = findX1(H)

m = length(H);
H0 = zeros(1,60);
X = zeros(1,length(H));
b = ones(1,60)*3;
f = ones(1,m);
for i=m:-1:1
    [tmp,Hj] = find(b>0);
    [y,x] = min(H0(Hj));%.*log(41-abs(Hj-20)).*log(41-abs(Hj-40)).*log(31-abs(Hj-30))
    x = Hj(x);
    if (i==m-1), x = 60; end;
    tmp = m:-1:1;
    for c=tmp
        if (f(c))
            break;
        end;
    end;
    f(c) = 0;
    X(c) = x;
    b(x) = b(x)-1;
    Y = Np(-37:37,H(c));
    for j=1:60
        t = 38-(x-j);
        if (t>=1) & (t<=75)
            H0(j) = H0(j) + Y(t);
        end;
    end;
    plot(1:60,H0);hold on;
    bar(x,1)
    pause;
end;

```

```

X = X+60*2;
function X = findX2(H)

m = length(H);
ro = 0.05;
k = 11;
H = [0 H'];
Ha = sum(H)/k;
D = ones(k,m);
%divide into k pieces
j=1;
for i=m+1:-1:1+1
    [a,b] = min(sum(H(D),2));
    for j=1:m
        if (D(b,j)==1)
            D(b,j) = i;
            break;
        end;
    end;
% bar(1:k,sum(H(D),2));
% pause
end;
%bar(((1:k)-1)*60/(k-1)+1,-sum(H(D),2)/1.4);
X = zeros(1,m);
b = ones(1,60)*2;
for i=1:k
    for t=1:m
        if (D(i,t)~=1)
            c = (i-1)*60/(k-1)+1;
            for j=[0 -1 1 -2 2 -3 3]
                if (c+j<=60 & c+j>=1)
                    if (b(c+j)>0)
                        b(c+j) = b(c+j) - 1;
                        X(D(i,t)-1) = c+j;
                        break;
                    end;
                end;
            end;
        end;
    end;
end;
X = X + 60*2;

```

```

function r = N(t,M)

```

```
t = (t)/(37.5)*3;
r = ZTF(t)*M;
```

```
function r = Np(t,M)
t = t/(37.5)*3;
r = ZTp(t)*M*(37.5/3);
function [r,d] = nNeed(H,X)
P = x2allH(H,X);
P = P;
m = 120 + 30;
tmp = need(H,X);
for r=1:100
    [P0,d] = doPass(P,r*185*0.92);
    if (P0(m)>=tmp(m)-20)
        break;
    end;
end;
figure;
plot(1:length(P),P,1:60*4,P0);%,1:60*4,tmp);
hold on;
%bar([38 38+60],ones(1,2)*8000);
```

```
function n = nNeedMin(H)
H = SZL(H,[0.85 0.7 0.75]);
X = findX2(H); P = x2allH(H,X); n = (sum(H)-0)/112.5*60/(185*0.92);
```

```
function r = SZL(H,L)
m = length(H);
for i=1:m
    if (H(i)<=85)
        r(i) = H(i) * L(1);
    elseif (H(i)<=215)
        r(i) = H(i) * L(2);
    else
        r(i) = H(i) * L(3);
    end;
end;
```

```
function r = X2allH(H,X)
X = X - 60*2;
r = zeros(1,60*4);
for i=1:length(H)
    tmp = N(-37:37,H(i));
```

```

r(X(i):(X(i)+74)) = r(X(i):(X(i)+74)) + tmp;
r(X(i)+75:(60*4)) = r(X(i)+75:(60*4))+ tmp(length(tmp));
end;

```

```

function r = X2allHp(H,X)
X = X - 60*2;

```

```

r = zeros(1,75+60);
for i=1:length(H)
    tmp = Np(-37:37,H(i));
    r(X(i):(X(i)+74)) = r(X(i):(X(i)+74)) + tmp;
end;

```

```

function ZT_init()
global ZT_F
global ZT_f
f = inline('(2*pi)^-0.5*exp(-t.^2/2)');
t = -4.99:0.01:0;
ZT_f = f(t);
for i=1:length(t)
    ZT_F(i) = sum(ZT_f(1:i).*0.01);
end;

```

```

function r = ZTF(t)
global ZT_F;
for i=1:length(t)
    if (t(i)>4.99)
        r(i) = 1;
    elseif (t(i)<-4.99)
        r(i) = 0;
    elseif (t(i)>0)
        r(i) = 1-ZTF(-t(i));
    else
        r(i) = ZT_F(round((t(i)+4.99)/0.01));
    end;
end;

```

```

function r = ZTp(t)
global ZT_f;
if (abs(t)>4.99)
    r = 0;
else
    r = ZT_f(round((-abs(t)+4.99)/0.01))*0.01;
end;

```

```
end;  
%result_flight2.m  
hold off;  
H=HA;  
X=findX2(H);  
H = [H' 68*1.4*ones(1,11)];  
X = [X 180:6:240];  
nNeed(H,X)  
hold on;  
[tmp,j] = sort(X);  
bar(tmp,H(j)/1.4 * 10);  
hold on;  
bar([30],ones(1,1)*1000);  
bar([90],ones(1,1)*3000);
```